

ThreatHunt: Detection and Response
Tool for Organizations Protection and Monitoring

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College of Information Technology Education
PHINMA University of Iloilo

In Partial Fulfillment
of the Requirements for the degree
Bachelor of Science in Information Technology

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Approval Sheet

In partial fulfillment of the requirements for the Bachelor of Science in Information Technology, this Capstone Project entitled ThreatHunt: Detection and Response Tool for Organizations Protection and Monitoring prepared and submitted by Permil Camilan, Jerome Fadrigo, Jemlyn Guirjen who are hereby recommended for corresponding final evaluation.

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ABSTRACT

In the modern and fast paced world today, data plays one of the most important roles for the organization to be functional. However the major concerns are how to protect those data in your organizations networks as it would be the primary target for unauthorized access. The need to prevent unauthorized access to data has become a major factor affecting the survival of organizations today due to the consequences that could arise when data falls into the wrong hands. This calls for the development of a new and improved detection, prevention and monitoring tool, creating the new data protection and prevention systems to mitigate those threats. This proposal makes a contribution through use of intrusion detection prevention systems to propose robust servers and network protection avoiding possible breach to organizations infrastructure. We propose this project to mitigate and solve this problem affecting almost all organizations that are using Windows systems to run and operate their networks and workspace especially with SSH servers.

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CHAPTER I

INTRODUCTION

In today's digital age, it is very important to protect sensitive information. Since organizations increasingly depend on digital platforms to save valuable information, they are open to cyberattacks and data breaches, which put their data at high risk. An Intrusion Detection System is very important in protecting against such attacks by tracking systems for malicious activity or unauthorized access in real-time, avoiding damage before it happens. Unfortunately, most organizations still use weak security measures, which leaves them open to cyberattacks. This study investigates how strengthening security with four key measures Intrusion Detection, Logging Failed Secure Shell Password Attempts, Firewall creation, and IP Blocking can enhance an organization's security. With these technologies, organizations can better protect sensitive information, identify threats early, and reduce the chances of breaches or unauthorized access.

Background of the Study

Intrusion Detection Prevention systems are essential tools that monitor and identify potential security threats in an organization's network in real-time. They play a crucial role in identifying unauthorized access, cyberattacks, and other malicious activities, helping organizations protect sensitive data and maintain system integrity. These systems provide early warnings, allowing for timely responses to threats and minimizing the risk of data breaches or financial loss.

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The importance of detection systems is amplified when combined with four key security measures: Intrusion Detection, Logging Failed Password Attempts for Secure Shell, Firewall creation, and IP Blocking. Logging failed password attempts helps detect brute-force attacks early. While IP blocking prevents known malicious addresses from accessing the network.

The detection prevention systems are vital for identifying and responding to potential security risks in real-time. By integrating them organizations can create a robust defense system capable of mitigating and preventing Secure Shell bruteforce attacks. This comprehensive approach ensures better protection of sensitive data, systems, and business operations.

Statement of the Problem

As cyber threats continue to grow in complexity and frequency, organizations are increasingly exposed to risks such as data breaches, unauthorized access, and cyberattacks. Despite the sensitivity and importance of the data they handle, including personal records and information, many companies rely on outdated, insufficient cybersecurity measures and unpatched software, leaving them exposed to potential threats. Many current security practices may lack real-time monitoring, effective logging, and robust firewall configurations, which are crucial for preventing unauthorized access and detecting suspicious activities. As a result, there is a risk of severe consequences, such as compromised data, financial loss, legal implications, and damage to the company's reputation.

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GENERAL OBJECTIVES

To develop ThreatHunt an intrusion detection prevention and prevention system.

SPECIFIC OBJECTIVES

Specifically, this project aims to achieve the following objective:

1. **SSH Intrusion detection:** By analyzing SSH failed login attempts.
2. **Firewall Rule Creation:** To enhance network security by controlling incoming and outgoing traffic, blocking unauthorized access, and preventing potential threats from reaching a system or network
3. **IP blocking:** If an attacker tries to break in from a specific internet address, you can block that address to stop them from getting in.
4. **Remote Connection Monitoring:** Monitor TCP connections of a computer to see where the computer is connecting.
5. **IP Lookup:** Perform identification of an IP from a successful connection to the host.

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SIGNIFICANCE OF THE STUDY

IT and Security Teams. The IT and security teams will benefit from the system by having a powerful tool to monitor and analyze logs, detect anomalies, and respond to security incidents.

Compliance Officers. Compliance officers will benefit from the system by having a solution that helps the company comply with relevant regulations and industry standards for security and data protection.

Employees. Employees will benefit from the system by having a secure working environment and protection from cyber threats that could compromise their personal data and productivity.

Regular Users. Regular users will benefit from the threat hunt system by having an increased confidence that their data is secure, which fosters trust in the company's ability to handle their sensitive information responsibly.

SCOPE AND LIMITATION

SCOPE:

The study focuses on development of the ThreatHunt for the system features include:

1. **SSH Intrusion Detection:** The system will evaluate the system's ability to detect potential intrusions and anomalies using the SSH log data or Windows Events.
2. **SSH Logs Monitoring:** The system will examine the

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system's log monitoring capabilities and its ability to provide valuable logs and data for forensic analysis.

3. **Firewall Creation:** The system will implement rules to block potential attacks and prevent unauthorized access to open services.

4. **Monitoring TCP Connection:** The system will monitor listening and outgoing network connections.

LIMITATION:

The delimitation of this study is as follows:

1. **Specific Threats:** The study will focus on specific security threats, such as brute force attacks on Secure Shell, other failed login attempts will not include other types of threats.

2. **Specific Operating System only :** The application is running only for Windows and no current support for other operating systems.

3. **Data Quality:** The system's ability to detect threats depends on the quality and amount of log data, which could lead to errors in detection.

4. **Non-Technical Factors:** The system doesn't address other important factors like staff training or organizational culture, which are also needed for strong security.

CHAPTER II

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LITERATURE REVIEW

In this study Vaisakhkrishnan et al.(2024) and Alabdulatif & Thilakarathne (2024) approach cybersecurity in healthcare from different angles, each emphasizing unique strategies to protect sensitive medical data. Vaisakhkrishnan et al. focus on a deep learning-based intrusion detection system (IDS) using Long Short-Term Memory (LSTM) networks to detect cyberthreats in IoMT environments, achieving excellent performance metrics, including 97% accuracy and 94% F1 score. Their work underscores the importance of automated, continuous threat detection to secure healthcare data. On the other hand, Alabdulatif & Thilakarathne advocate for a multi-layered approach to security that includes user authentication, regular system updates, and advanced security technologies like firewalls and encryption. They also highlight the importance of cybersecurity training and threat intelligence to proactively address vulnerabilities. While Vaisakhkrishnan et al. focus on deep learning for real-time intrusion detection, Alabdulatif & Thilakarathne emphasize preventive measures and strategic preparedness, showcasing a contrast in their methodologies, one being reactive and the other proactive.

Similarly, Algethami & Alshamrani (2024) and Sun et al. (2024) propose advanced systems for intrusion detection in healthcare environments but use different technological approaches. Algethami & Alshamrani present a hybrid deep learning system that combines Gated Recurrent Unit (GRU), Bidirectional Long Short-Term Memory (BLSTM), and Artificial Neural Networks (ANN) to secure IoHT devices, achieving remarkable results with 100% accuracy and precision. Their use of cutting-edge AI methods

highlights the effectiveness of deep learning in providing

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accurate and efficient threat detection. In contrast, Sun et al. (2024) combine Particle Swarm Optimization (PSO) with AdaBoost to detect malware related records in health app platforms, achieving high recall (0.966667) and superior performance in comparison to alternative techniques. While both studies focus on machine learning-based intrusion detection, Algethami & Alshamrani's work emphasizes hybrid deep learning models for high accuracy, while Sun et al. focus on the synergy of optimization and boosting algorithms to enhance detection capabilities in health app platforms, reflecting different approaches in improving the reliability of cybersecurity .

Judith et al. (2023) and Algethami & Alshamrani (2024) both focus on the security of the Internet of Medical Things (IoMT) but tackle the problem from different perspectives. Judith et al. address the dynamic nature of cyberattacks in IoMT networks, specifically focusing on man-in-the-middle attacks and utilizing multi-layer perceptron (MLP) and principal component analysis (PCA) to improve detection, achieving 96.39% accuracy. Their work emphasizes the need for scalable solutions in addressing evolving cyber threats. In contrast, Algethami & Alshamrani take a more comprehensive approach by integrating various deep learning techniques such as GRU, BLSTM, and ANN to secure IoHT environments, achieving a perfect detection score in binary classifications. While both works acknowledge the critical role of machine learning in addressing cybersecurity threats in healthcare, Judith et al. prioritize efficient attack classification and feature reduction, while Algethami & Alshamrani explore a hybrid deep learning approach for a broader range of cybersecurity threats, reflecting differing priorities in securing healthcare data.

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REFERENCES

- Vaisakhkrishnan, K., Ashok, G., Mishra, P., & Kumar, T. G. (2024). Guarding Digital Health: deep learning for attack detection in Medical ISIoT. *Procedia Computer Science*, 235, 2498-2507.
- Alabdulatif, A., & Thilakarathne, N. N. (2024). A novel cloud-enabled cyber threat hunting platform for evaluating the cyber risks associated with smart health ecosystems. *Applied Sciences*, 14(20), 9567.
- Algethami, S. A., & Alshamrani, S. S. (2024). A deep learning-based framework for strengthening cybersecurity in internet of health (IoHT) environments. *Applied Sciences*, 14(11), 4729
- Sun, Z., An, G., Yang, Y., & Liu, Y. (2024). Optimized machine learning enabled intrusion detection 2 systems for the internet of medical things. *Franklin Open*, 6, 100056.
- medical things devices. *Engineering Proceedings*, 59(1), 139.

OVERVIEW

Intrusion detection in companies is vital for protecting sensitive data from cyberthreats, and recent studies have explored various approaches to tackle this issue. Vaisakhkrishnan et al. (2024) focus on a deep learning-based intrusion detection system using Long Short-Term Memory (LSTM) networks for real-time threat detection in the Internet of Medical Things (IoMT), achieving high accuracy and F1 scores. In contrast, Alabdulatif & Thilakarathne (2024) advocate for a multi-layered, proactive approach involving user authentication, system updates, encryption, and cybersecurity training to prevent attacks. Other research, like that of Algethami &

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Alshamrani (2024), uses a hybrid deep learning system combining Gated Recurrent Unit (GRU), Bidirectional LSTM (BLSTM), and Artificial Neural Networks (ANN) to secure IoHT devices with perfect detection accuracy. Sun et al. (2024) combine Particle Swarm Optimization (PSO) with AdaBoost to detect malware in health app platforms, achieving high recall. Judith et al. (2023) address man-in-the-middle attacks in IoMT networks using multi-layer perceptron (MLP) and principal component analysis (PCA), focusing on scalable solutions, while Algethami & Alshamrani use deep learning techniques for broader cybersecurity threat detection. These studies highlight a variety of strategies, from real-time deep learning models to proactive measures, all aiming to enhance cybersecurity in healthcare environments.

TRADITIONAL VS MODERN SYSTEM

In terms of company cybersecurity, the study contrasts conventional and contemporary intrusion detection systems (IDS). To proactively address vulnerabilities and stop assaults, traditional intrusion detection systems (IDS), such as the multi-layered security strategy proposed by Alabdulatif & Thilakarathne (2024), rely on preventive measures including user authentication, firewalls, encryption, and system updates. This approach, meanwhile, could not be as flexible in the face of emerging, invisible dangers. Modern IDS, on the other hand, employ sophisticated machine learning techniques like Long Short-Term Memory (LSTM) and hybrid models (e.g., GRU, BLSTM, ANN) for real-time detection of evolving cyber threats. Examples of these systems are the deep learning-based systems suggested by Vaisakhkrishnan et al. (2024) and Algethami & Alshamrani (2024). These systems are very good at spotting new assaults

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because they are constantly learning from data and attain great accuracy and precision (e.g., Vaisakhkrishnan et al. reached 97% accuracy). Modern IDS provides adaptive, reactive detection that can swiftly identify and neutralize complex cyberthreats in healthcare settings, whereas traditional IDS concentrates on prevention and readiness.

TECHNOLOGY USE

The technologies used in this topic mainly use advanced machine learning and deep learning for cybersecurity in healthcare, especially for IoMT and IoHT. Key technologies include:

1.Long Short-Term Memory (LSTM) Networks: Vaisakhkrishnan et al. (2024) use LSTM networks for real-time intrusion detection in IoMT environments. LSTM networks are good at spotting patterns and anomalies over time, making them perfect for constantly monitoring healthcare data.

2.Gated Recurrent Units (GRU) and Bidirectional LSTM (BLSTM): Algethami & Alshamrani (2024) use hybrid deep learning models to protect IoHT devices. These models analyze data over time and detect threats accurately and quickly in real-time.

3.Artificial Neural Networks (ANN): Algethami & Alshamrani use GRU and BLSTM together for strong threat detection in healthcare settings.

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4. Particle Swarm Optimization (PSO) and AdaBoost: Sun et al. (2024) use PSO to select important features and AdaBoost to improve detecting malware in health apps. These methods make the detection is more accurate and performs better than traditional approaches.

5. Multi-Layer Perceptron (MLP) and Principal Component Analysis (PCA): Judith et al. (2023) focus on detecting man-in-the-middle attacks using MLP and PCA to reduce features, which helps improve detection efficiency in IoMT networks.

These advanced technologies, including deep learning, machine learning optimization algorithms, and feature reduction techniques, are integral to modern cybersecurity systems in healthcare, enabling more accurate, adaptive, and efficient detection of cyber threats.

CASE STUDIES

The case studies showcase different approaches to enhancing cybersecurity in healthcare environments, particularly focusing on the Internet of Medical Things (IoMT) and the Internet of Healthcare Things (IoHT).

1. Vaisakhkrishnan et al. (2024) presents a deep learning-based intrusion detection system (IDS) using Long Short-Term Memory (LSTM) networks to detect cyber threats in IoMT environments. It shows the system's effectiveness with 97% accuracy and a 94% F1 score. This highlights the importance of automated, real-time monitoring to detect threats and protect healthcare data,

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offering a reactive approach to cybersecurity.

2. **Alabdulatif & Thilakarathne (2024)** They take a different approach by supporting a multi-layered, proactive security strategy. Their case study focuses on using measures like user authentication, system updates, encryption, and cybersecurity training. This highlights the value of preventing threats before they happen, creating a secure and resilient healthcare environment.

3. **Algethami & Alshamrani (2024)** They propose a hybrid deep learning system combining GRU, BLSTM, and ANN to protect IoHT devices. Their case study shows outstanding results with 100% accuracy and precision, highlighting how these combined techniques provide accurate and efficient threat detection.

4. **Sun et al. (2024)** They combine Particle Swarm Optimization (PSO) with AdaBoost to detect malware in health apps. Their case study shows high recall (0.966667) and better performance than other methods, highlighting how combining optimization and boosting improves malware detection in healthcare apps.

5. **Judith et al. (2023)** They focus on detecting man-in-the-middle attacks in IoMT networks using MLP and PCA, achieving 96.39% accuracy. Their case study shows the need for scalable solutions to handle changing cyberattacks. In contrast, Algethami & Alshamrani use a hybrid deep learning system (GRU, BLSTM, and ANN) to protect IoHT environments, achieving perfect detection accuracy.

CHAPTER III

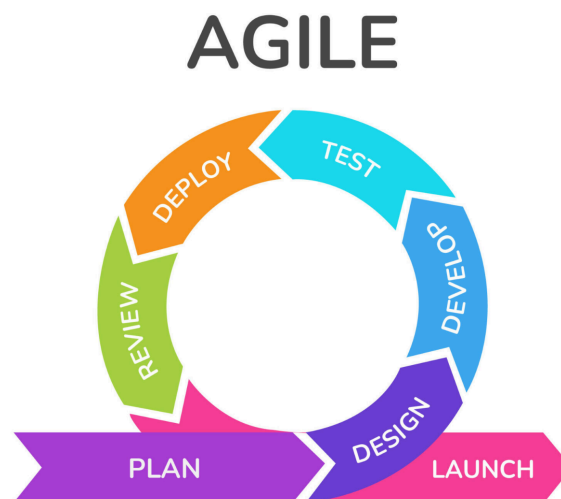
METHODOLOGY

This chapter explains the research process for developing and evaluating ThreatHunt, which focuses on improving security for organizations by using Intrusion Detection and Prevention using firewall logs , Logging Failed Password Attempts for Secure Shell, Firewall Creation, and IP Blocking.

RESEARCH DESIGN

This study employed a developmental research design focusing on the design, development, and evaluation of the proposed system. Descriptive methods were also utilized to gather data from the client by presenting the system and obtaining feedback to identify requirements and to further improve its functionality.

MODEL USE



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The project uses the Agile Development Model, which is flexible and emphasizes teamwork, adaptability, and continuous improvement. The development process is divided into small phases, allowing regular testing and updates to the system. This approach ensures that new security features and improvements can be integrated efficiently as the system evolves.

REQUIREMENTS PHASE

In the Requirements Phase, the goal is to define the user needs and features for the system. Key user needs include detecting unauthorized access (intrusions), tracking failed SSH login attempts to prevent brute-force attacks, and allowing users to block malicious IP addresses. The system should include an Intrusion Detection System (IDS) for real-time threat detection, a logging system for failed SSH login attempts, firewall improvements, and an IP blocking feature.

DESIGN PHASE

In the Design Phase, the system will be built with different parts to tackle specific security issues. First, an Intrusion Detection System (IDS) will monitor network traffic and server activity in real time using event security log to detect threats. The logging system will track failed SSH login attempts, security events, and intrusions, ensuring that logs are stored securely and timestamped for easy review. The firewall management module will allow users to create their own rules to block harmful traffic and unauthorized access. The IP blocking system will identify and block harmful IP addresses based on suspicious activities or attempts.

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DEVELOPMENT PHASE

In the Development Phase, the system is built in multiple steps. Part 1 focuses on creating the Intrusion Detection System (IDS), which will monitor network traffic for failed unauthorized SSH access and log alerts in real time. Part 2 will develop the logging module to track failed login attempts and display alerts for repeated failures. Part 3 will create the IP blocking feature, which allows users to block IP addresses linked to malicious IP that can be found in the logs. Part 4 will improve the firewall, allowing the user to create their own firewall rules. Finally, Part 5 will bring everything together, integrating all components into the system and testing basic functions to ensure everything works correctly.

TESTING PHASE

In the Testing Phase, different types of testing will be done to make sure the system works well. Unit testing will check each part of the system, like the IDS, logging, firewall, and IP blocking. Integration testing will ensure all the parts work together smoothly. Performance testing will check if the system can handle high traffic and logs without delays in real-time detection. Lastly, user acceptance testing will be done with the client to get feedback on how the system works in real-world situations.

DEPLOYMENT PHASE

In the Deployment Phase, the application will first be deployed to a staging environment for final testing. This setup simulates real conditions while keeping the system separate from

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the live network. The focus will be on making sure real-time monitoring, logging, and alerts work correctly without causing disruptions. After successful tests in the staging environment, the system will be moved to monitoring to track the system's performance, such as detection rates and response times, and make any needed adjustments based on how it performs in the testing environment.

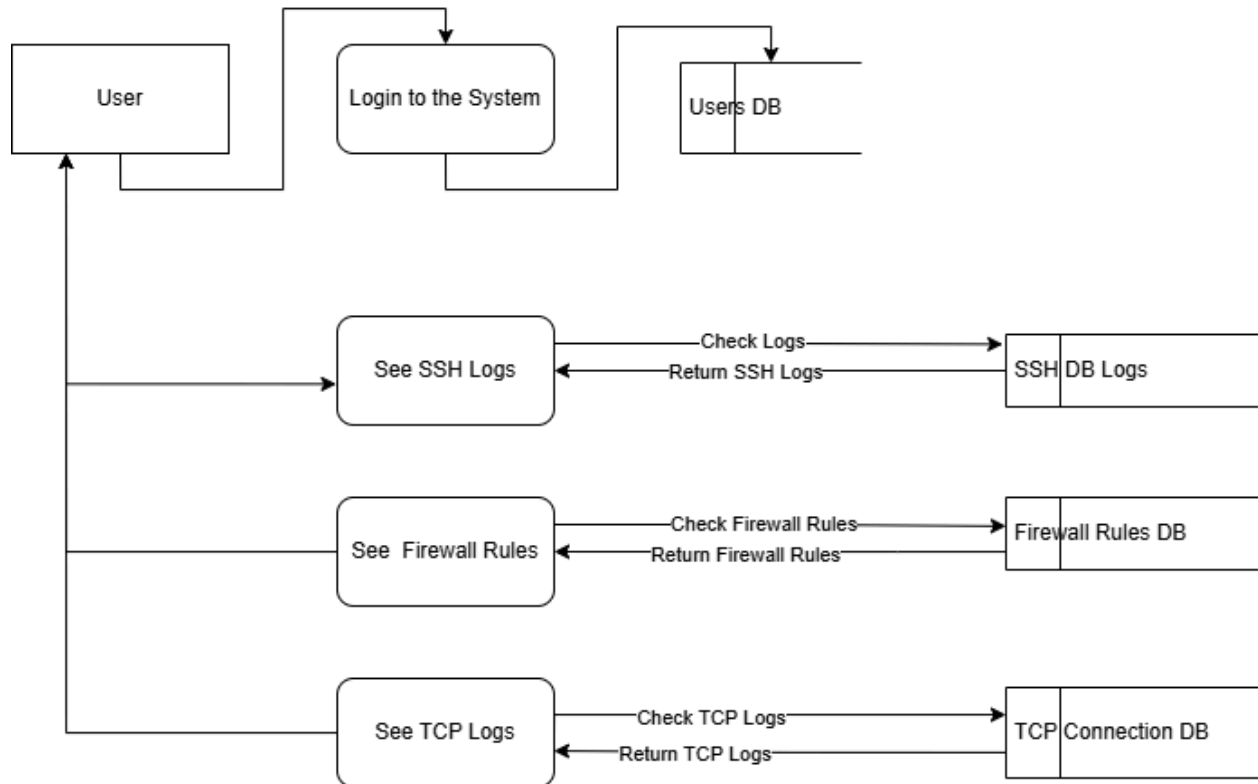
REVIEW PHASE

In the Review Phase, the performance of the application will be assessed after deployment. Key metrics like detection rates, false positives, and response times will be reviewed to check how well the system is working. Feedback will be collected from our clients. Continuous monitoring will also be done to spot new threats and ensure the system stays effective as cybersecurity risks change.

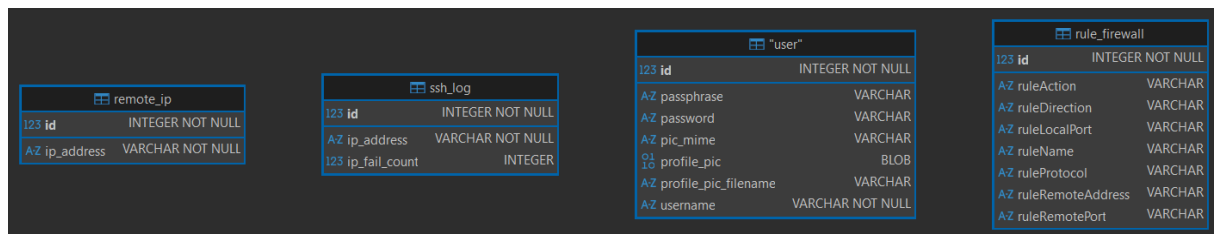
This Agile approach makes the development of the ThreatHunt flexible and ongoing, allowing for continuous improvements based on performance data and user feedback. Each phase requirement, design, development, testing, deployment, and review builds on the previous one to create a secure and reliable system that protects sensitive data from cyber threats.

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DATA FLOW DIAGRAM



ENTITY RELATIONSHIP DIAGRAM



TECHNOLOGY STACK

The project uses this technology stack to deliver satisfaction to the client, ensuring to meet their expectations and enhance , help and protect their organization or business. The system is a combination of web-based programming languages and firewall modules as being listed below:

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- Languages: Python, Powershell, Javascript
- Database: SQLite
- Tools: VSCode, OpenSSH-Server
- Environment: Windows Server, Linux Server
- Framework: Flask, Websocket

DATA GATHERING

This survey was conducted to collect feedback from users regarding their current methods of monitoring network security and managing server access. It also asks about what users want or expect from the new system called "ThreatHunt: Detection and Response Tool for Organizations Protection and Monitoring", the answers from this survey helped in creating a system that was functional, secure, and specifically suited to the needs of its technical users in identifying and mitigating network threats.

Demographic Information

1. What is your primary role in the IT organization?

- System Administrator
- Security Analyst / SOC Engineer
- Network Engineer
- IT Manager / Consultant

2. How many years of experience do you have in network security or server management?

- Less than 1 year
- 1-3 years
- 4-6 years

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- More than 6 years

3. Which operating environment do you primarily manage?

- Windows Server
- Linux/Unix
- Hybrid (Both Windows & Linux)
- Cloud-based (AWS/Azure/GCP)

SSH Intrusion Detection & Logs

4. How accurately does the system detect failed SSH login attempts?

- Highly Accurate
- Accurate
- Occasionally Inaccurate
- Inaccurate

5. How useful is the categorization of "Brute-Force Attacks" for your daily monitoring?

- Very Useful
- Useful
- Somewhat Useful
- Not Useful at all

6. How would you rate the clarity of the SSH log entries (Timestamps, IPs, Usernames)?

- Excellent (Very easy to read)
- Good (Clear enough)

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- Fair (Somewhat cluttered)
- Poor (Difficult to interpret)

7. Does the system provide sufficient data for forensic analysis after a threat is detected?

- Strongly Agree
- Agree
- Disagree
- Strongly Disagree

Firewall & TCP Connection Monitoring

8. How intuitive is the process of creating new firewall rules in the system?

- Very Intuitive
- Intuitive
- Complex
- Very Difficult

9. How effective is the system at blocking unauthorized access once a rule is set?

- Highly Effective
- Effective
- Inconsistent
- Ineffective

10. How would you rate the real-time display of active TCP connections?

- Very Responsive

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- Responsive
- Delayed
- Non-responsive

11. Does the monitoring of both listening and outgoing ports meet your security needs?

- Exceeds Needs
- Meets Needs
- Below Needs
- Does Not Meet Needs

System Performance and Usability

12. How would you rate the speed of the system when analyzing Windows Event logs?

- Very Fast
- Satisfactory
- Slow
- Very Slow

13. How user-friendly is the overall dashboard interface?

- Very User-friendly
- User-friendly
- Complex
- Hard to Navigate

14. How satisfied are you with the system's ability to handle multiple events simultaneously?

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- Very Satisfied
- Satisfied
- Unsatisfied
- Very Unsatisfied

15. What is your preferred platform for monitoring these security alerts?

- Desktop Application
- Web-based Dashboard
- Mobile/Tablet View
- Command Line Interface (CLI)

Reliability and Future Implementation

16. How important is the "Dynamic Management" (modifying rules on the fly) feature to you?

- Critical
- Important
- Neutral
- Not Important

17. What is your primary concern when using this intrusion detection system?

- System Resource Usage (CPU/RAM)
- Complexity of Setup
- Accuracy of Alerts
- Integration with other tools

18. How likely are you to recommend ThreatHunt as a primary

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security tool?

- Very Likely
- Likely
- Unlikely
- Very Unlikely

19. How much training do you feel is required to operate the Firewall module effectively?

- No training needed
- Minimal training
- Moderate training
- Extensive training

20. Overall, does the system successfully meet the objective of securing SSH and Network traffic?

- Fully Meets Objectives
- Partially Meets Objectives
- Needs Significant Improvement
- Fails Objectives

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CHAPTER IV

Result and Discussion

This chapter presents the results of the system development and evaluation. It discusses how each feature of the system performed during testing and how it met the objectives stated in the study. The results are analyzed and interpreted in relation to the system's scope and limitations.

Task Manager: Gantt Chart/Project Timetable - ThreatHunt: Detection and Response Tool for Organizations Protection and Monitoring

Week	Task	Programmer	UI / UX	Documenter
1	Project Planning & Requirements Gathering	✓	✓	✓
2	Research (SSH Logs & Windows Events) & Design	✓	✓	✓
3	Use Case & DFD (Data Flow Diagram) Creation			✓
4	Database Design & Schema Creation	✓	✓	✓
5	UI Wireframes & Dashboard Mockups		✓	
6	Frontend Development (Monitoring Dashboard)	✓	✓	
7	Backend Development (Detection & Firewall Logic)	✓		

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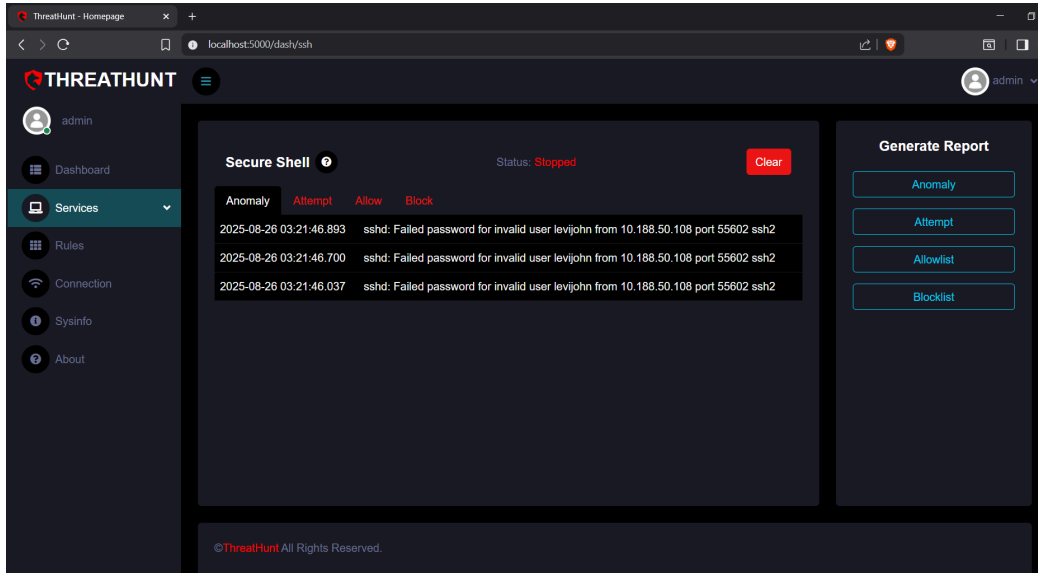
8	Integration of Frontend and Backend	✓	✓	
9	System Testing & Bug Fixing (Debugging)	✓	✓	✓
10	User Testing & Feedback Collection			✓
11	Final Revisions (Code + UI + Content)	✓	✓	✓
12	Documentation Finalization & Formatting			✓
	Final Project Submission	✓	✓	✓

System's Modules

1. SSH Intrusion Detection

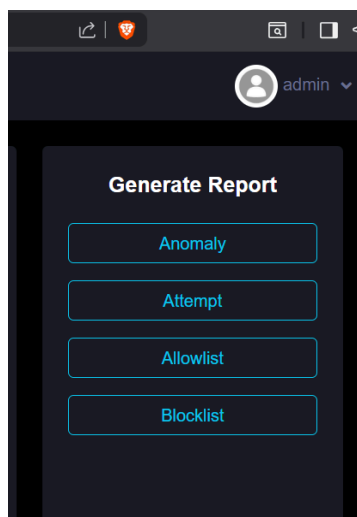
The system successfully detected multiple failed SSH login attempts, categorizing them as potential brute-force attacks. The detection module analyzed the Windows Event logs and SSH log data, identifying repeated access attempts from the same IP address within a short period.

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2.SSH Logs Monitoring

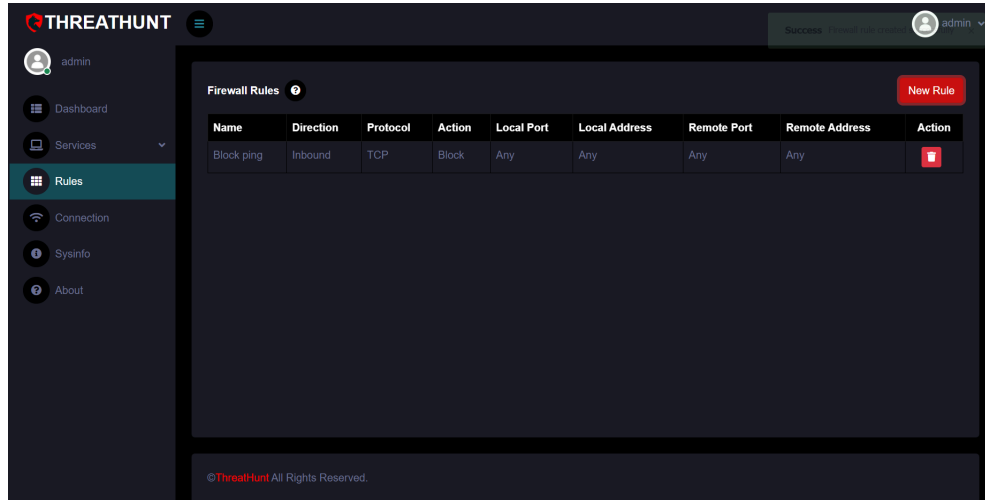
The system was able to capture and display SSH log entries containing essential information such as timestamps, usernames, source IPs, and login status. These logs were stored for future forensic analysis, allowing administrators to trace suspicious activities.



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3. Firewall Creation

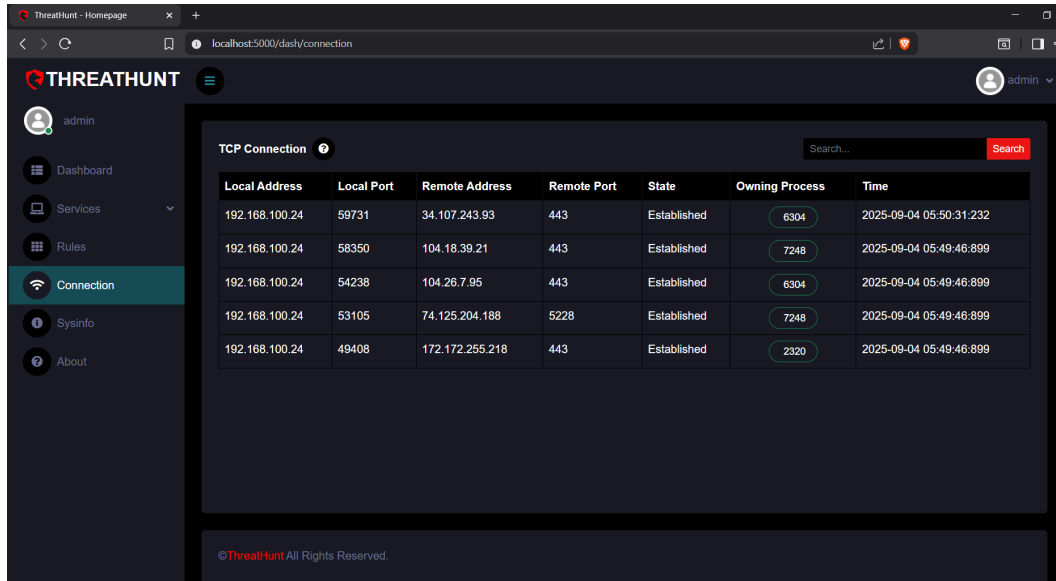
The system successfully allows users to create their own firewall rules tailored for their own use case. The system implements a dynamic firewall management module capable of adding or modifying firewall rules.



4. Monitoring TCP Connections

The system successfully monitored both listening and outgoing TCP connections, displaying active ports and connected IP addresses. Administrators could view these connections in real time, helping detect unusual or unauthorized network activity.

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The screenshot displays the ThreatHunt web application interface. The left sidebar contains navigation links: Dashboard, Services, Rules, Connection (highlighted), Sysinfo, and About. The main content area is titled 'TCP Connection' and features a search bar. Below the search bar is a table listing active TCP connections. The table has columns for Local Address, Local Port, Remote Address, Remote Port, State, Owning Process, and Time. The data shows five established connections from the local IP 192.168.100.24 to various remote addresses, all with the state 'Established' and associated with processes 6304, 7248, or 2320.

Local Address	Local Port	Remote Address	Remote Port	State	Owning Process	Time
192.168.100.24	59731	34.107.243.93	443	Established	6304	2025-09-04 05:50:31.232
192.168.100.24	58350	104.18.39.21	443	Established	7248	2025-09-04 05:49:46.899
192.168.100.24	54238	104.26.7.95	443	Established	6304	2025-09-04 05:49:46.899
192.168.100.24	53105	74.125.204.188	5228	Established	7248	2025-09-04 05:49:46.899
192.168.100.24	49408	172.172.255.218	443	Established	2320	2025-09-04 05:49:46.899

System's Features

1. **SSH Intrusion Detection:** The system will evaluate the ThreatHunt system's ability to detect potential intrusions and anomalies using the SSH log data or Windows Events.
2. **SSH Logs Monitoring:** The system will examine the system's log monitoring capabilities and its ability to provide valuable logs and data for forensic analysis.
3. **Firewall Creation:** The system will implement rules to block potential attacks and prevent unauthorized access to open services.
4. **Monitoring TCP Connection:** The system will monitor listening and outgoing network connections.

Implementation Challenges

Implementing the ThreatHunt is a bit complicated since the program solely runs in the Windows operating system environment

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and it only targets the SSH service in the Windows system. Also the program requires an administrator privilege to run and function properly and requires resource usage in the system. These implementation challenges

User Feedback

Client expressed greater confidence in the security of their network, pointing to its proactive detection of SSH intrusion attempts and real-time TCP connection monitoring. Their comprehension of system vulnerabilities and defensive configuration was improved by practical experience with SSH log analysis and firewall rule creation. The client mentioned that

setting up customized firewall rules and deciphering log data for precise threat detection presented some initial difficulties. In general, comments emphasized the system's instructional value, intuitive layout, and usefulness in enhancing network security and blocking unwanted access in security scenario simulations.

Comparison to other System

Compared to other security and monitoring tools, this system provides a more streamlined and user-friendly approach to detecting SSH intrusions and managing firewall rules. While well-known platforms such as Snort, Suricata, and Splunk offer extensive capabilities for large-scale network monitoring and threat analysis, they often require advanced technical knowledge and complex setup procedures. In contrast, this project emphasizes simplicity and accessibility, allowing users to easily monitor SSH activities, detect suspicious login attempts,

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and configure firewall rules without the need for specialized expertise.

Another key advantage is its executable design, which allows it to run efficiently on Windows systems without requiring packages to be installed or additional software licenses. This makes it particularly suitable for small organizations, individual users who need reliable network protection without the complexity of enterprise solutions. However, unlike cross-platform tools such as Wireshark, the system currently operates only on Windows and focuses primarily on SSH-related threats rather than providing comprehensive network threat coverage. Despite this limitation, its combination of real-time monitoring, IP blocking, and customizable security rules offers a balanced solution that enhances cybersecurity awareness and supports practical defense against unauthorized access.

Project Structure

```
├─ app.py
└─ main
    ├── __init__.py
    ├── instance
    │   ├── flask_session
    │   │   └─ efe08b5ff51a6b44fa056ee99add1453
    │   └─ user.db
    ├── models.py
    ├── modules
    │   ├── connectionModule
    │   │   ├── connectionHandling.py
    │   │   ├── connection_tcp.py
    │   │   └─ connectionThread.py
```

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```
|   └─ dashModule
|   |   └─ dashFetchContent.py
|   └─ ruleModule
|   |   └─ ruleFirewallHandle.py
|   └─ sshModule
|   |   └─ convertTimestamp.py
|   |   └─ sshFirewallHandling
|   |       └─ ssh_firewall_grab_rule.py
|   |       └─ ssh_firewall_rule_handling.py
|   |       └─ ssh_firewall_rule_route.py
|   |   └─ ssh_logger_websocket.py
|   |   └─ ssh_status_emitter.py
|   └─ sysinfoModule
|   |   └─ sysinfo.py
|   └─ userProfileModule
|       └─ getUserInfo.py
|       └─ updateUserProfile.py
|       └─ uploadProfile.py
└─ routes
    └─ auth.py
    └─ homeRoute.py
└─ static
    └─ css
        └─ all.min.css
        └─ bootstrap.min.css
        └─ leaflet.css
        └─ style.css
        └─ tabulator.min.css
    └─ images
        └─ default-image.jpg
        └─ logo-ico.png
        └─ members
```

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```
| | | └─ 1.jpg
| | | └─ 2.jpg
| | | └─ 3.jpg
| | └─ js
| | | └─ auth.js
| | | └─ connection
| | | | └─ connection.js
| | | └─ dashboard
| | | | └─ dash.board.js
| | | | └─ dash.chart.js
| | | | └─ dash.fetch.js
| | | └─ jquery-3.7.1.min.js
| | | └─ leaflet.js
| | | └─ main.js
| | | └─ profile
| | | | └─ preview.profile.js
| | | | └─ update.profile.js
| | | | └─ upload.profile.js
| | | └─ rule
| | | | └─ rule.fetch.firewall.js
| | | └─ socket.io.min.js
| | | └─ ssh
| | | | └─ ssh.allow.block.js
| | | | └─ ssh.clear.log.js
| | | | └─ ssh.content.download.js
| | | | └─ ssh.websocket.js
| | | └─ sysinfo
| | | | └─ sysinfo.js
| | | └─ tabulator.min.js
| | └─ lib
| | | └─ bootstrap.bundle.min.js
| | | └─ chart
```

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```
|      |      └─ chart.min.js
|      |      └─ easing
|      |      └─ easing.js
|      |      └─ easing.min.js
|      └─ html2pdf.bundle.min.js
|      └─ iziToast
|      |      └─ css
|      |      |      └─ iziToast.css
|      |      └─ js
|      |          └─ iziToast.js
|      └─ owlcarousel
|      |      └─ assets
|      |      |      └─ ajax-loader.gif
|      |      |      └─ owl.carousel.css
|      |      |      └─ owl.carousel.min.css
|      |      |      └─ owl.theme.default.css
|      |      |      └─ owl.theme.default.min.css
|      |      |      └─ owl.theme.green.css
|      |      |      └─ owl.theme.green.min.css
|      |      |      └─ owl.video.play.png
|      |      └─ owl.carousel.js
|      |      └─ owl.carousel.min.js
|      └─ tempusdominus
|      |      └─ css
|      |      |      └─ tempusdominus-bootstrap-4.css
|      |      |      └─ tempusdominus-bootstrap-4.min.css
|      |      └─ js
|      |          └─ moment.min.js
|      |          └─ moment-timezone.min.js
|      |          └─ tempusdominus-bootstrap-4.js
|      |          └─ tempusdominus-bootstrap-4.min.js
└─ templates
```

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```
|— components
|   |— footer.html
|   |— head.html
|   |— javascript.html
|   |— navbar.html
|   └— not-signin.html
|— dash-about.html
|— dash-board.html
|— dash-connection.html
|— dash-profile.html
|— dash-rules.html
|— dash-ssh.html
|— dash-sysinfo.html
|— index.html
|— not-signin.html
|— partials
|   └— sidebar.html
└— signin.html
```

Code Impression Management

app.py

```
from gevent import monkey
monkey.patch_all()

import geventwebsocket
from main import create_app, socketio

app = create_app()

if __name__ == "__main__":
    socketio.run(app, host="0.0.0.0", port=5000,
debug=True, use_reloader=False)
```


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main/___init___py

```
1  import os
2  import sys
3  from flask import Flask, session, Blueprint, jsonify
4  from flask session import Session
5  from flask socketio import SocketIO
6  from .models import db
7
8  # Initialize extensions globally
9  server_session = Session()
10 socketio = SocketIO(
11     cors_allowed_origins="*",
12     async_mode="gevent",
13     logger=True,
14     engineio_logger=True
15 )
16
17
18 def create_app():
19     app = Flask(__name__)
20
21     @app.template_filter('obfuscate')
22     def obfuscate_filter(filename):
23         if app.debug:
24             return f"js/{filename}"
25         return f"dist/{filename}"
26
27     # Detect if running from PyInstaller .exe
28     if getattr(sys, "frozen", False):
29         # If frozen, use directory of the executable
30         basedir = os.path.dirname(sys.executable)
31     else:
32         basedir = os.path.abspath(os.path.dirname(__file__))
33
34     # Persistent instance paths
35     instance_path = os.path.join(basedir, "instance")
36     os.makedirs(instance_path, exist_ok=True)
37     session_dir = os.path.join(instance_path, "flask_session")
38     os.makedirs(session_dir, exist_ok=True)
```

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```
40 app.config["SECRET_KEY"] = "SFSFSFdsfsfe2326^@&^7"
41
42 app.config["SESSION_TYPE"] = "filesystem"
43 app.config["SESSION_FILE_DIR"] = session_dir
44 app.config["SESSION_PERMANENT"] = False
45 app.config["SESSION_USE_SIGNER"] = True
46 app.config["SESSION_COOKIE_HTTPONLY"] = True
47 app.config["SESSION_COOKIE_SECURE"] = False
48 app.config["SESSION_COOKIE_SAMESITE"] = "Lax"
49 app.config["SESSION_REFRESH_EACH_REQUEST"] = True
50
51 server_session.init_app(app)
52
53 @app.context_processor
54 def inject_user():
55     return dict(
56         logged_in=session.get("logged_in"), username=session.get("username")
57     )
58
59 # Database setup (persistent)
60 db_path = os.path.join(instance_path, "user.db")
61 app.config["SQLALCHEMY_DATABASE_URI"] = f"sqlite:/// {db_path}"
62 app.config["SQLALCHEMY_TRACK_MODIFICATIONS"] = False
63 db.init_app(app)
64
65 with app.app_context():
66     db.create_all()
67
68 # Blueprints & Modules
69 from .routes.auth import auth_bp
70 from .routes.homeRoute import homeRoute_bp
71 from .modules.dashModule.dashFetchContent import dashboard_view_bp
72 from .modules.userProfileModule.getUserInfo import get_User_profile_bp
73 from .modules.userProfileModule.updateUserProfile import update_profile_bp
74 from .modules.userProfileModule.uploadProfile import myprofile_bp
75 from .modules.ruleModule.ruleFirewallHandle import rules_firewall_handler_bp
76 from .modules.connectionModule import connection_tcp
77 from .modules.connectionModule.connectionHandling import connection_handling_bp
```

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```
78 from .modules.sysinfoModule.sysinfo import sysinfo_blueprint
79 from main.modules.sshModule import ssh_status_emitter, ssh_logger_websocket
80 from main.modules.sshModule.sshFirewallHandling.ssh_firewall_rule_route import (
81     ssh_firewall_handler,
82 )
83 from main.modules.sshModule.sshFirewallHandling.ssh_firewall_grab_rule import (
84     ssh_firewall_get_allow_block_rule_bp,
85 )
86 from .modules.connectionModule.connectionThread import start_connection_threads
87
88 app.register_blueprint(auth_bp, url_prefix="/")
89 app.register_blueprint(homeRoute_bp, url_prefix="/dash")
90
91 api_bp = Blueprint("api", __name__)
92
93 @api_bp.before_request
94 def require_login():
95     if not session.get("logged_in"):
96         return jsonify({"error": "Unauthorized"}), 401
97
98 api_bp.register_blueprint(dashboard_view_bp, url_prefix="/board")
99 api_bp.register_blueprint(ssh_firewall_handler, url_prefix="/ssh")
100 api_bp.register_blueprint(ssh_firewall_get_allow_block_rule_bp, url_prefix="/ssh")
101 api_bp.register_blueprint(connection_handling_bp, url_prefix="/connection")
102 api_bp.register_blueprint(rules_firewall_handler_bp, url_prefix="/rule")
103 api_bp.register_blueprint(sysinfo_blueprint)
104 api_bp.register_blueprint(myprofile_bp, url_prefix="/profile")
105 api_bp.register_blueprint(get_User_profile_bp)
106 api_bp.register_blueprint(update_profile_bp, url_prefix="/profile")
107 app.register_blueprint(api_bp, url_prefix="/api")
108
109 socketio.init_app(app)
110
111 if not os.environ.get('WERKZEUG_RUN_MAIN') == 'true':
112     if not hasattr(app, 'log_thread_started'):
113         ssh_logger_websocket.start_log_thread(app)
114         app.log_thread_started = True
115
116 return app
117
```

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models.py

```
1  from flask sqlalchemy import SQLAlchemy
2
3  db = SQLAlchemy()
4
5
6  class User(db.Model):
7      id = db.Column(db.Integer, primary_key=True)
8      username = db.Column(db.String(50), unique=True, nullable=False)
9      password = db.Column(db.String(50))
10     passphrase = db.Column(db.String(50))
11     profile_pic = db.Column(db.LargeBinary)
12     pic_mime = db.Column(db.String(50))
13     profile_pic_filename = db.Column(db.String(100))
14
15
16 class ruleFirewall(db.Model):
17     id = db.Column(db.Integer, primary_key=True)
18     ruleName = db.Column(db.String(50))
19     ruleDirection = db.Column(db.String(50))
20     ruleProtocol = db.Column(db.String(50))
21     ruleAction = db.Column(db.String(50))
22     ruleLocalPort = db.Column(db.String(50))
23     ruleRemotePort = db.Column(db.String(50))
24     ruleRemoteAddress = db.Column(db.String(50))
25
26
27 class RemoteIP(db.Model):
28     id = db.Column(db.Integer, primary_key=True)
29     ip_address = db.Column(db.String(50), unique=True, nullable=False)
30
31
32 class SSHLog(db.Model):
33     id = db.Column(db.Integer, primary_key=True)
34     ip_address = db.Column(db.String(50), unique=True, nullable=False)
35     ip_fail_count = db.Column(db.Integer, default=0)
```

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main/modules/connectionModule/connection_tcp.py

```
1  import subprocess
2  import json
3  import re
4  import ipaddress
5  from flask import current_app
6  from main import socketio
7  from main.models import db, RemoteIP
8  from datetime import datetime
9
10 connection_timestamps = {}
11
12
13 def get_tcp_connections():
14     try:
15         result = subprocess.run(
16             ["ss", "-tunap"],
17             capture_output=True,
18             text=True
19         )
20
21         if result.returncode != 0:
22             return None, result.stderr
23
24         lines = result.stdout.splitlines()
25         connections = []
26
27         for line in lines[1:]:
28             parts = line.split()
29             if len(parts) < 6:
30                 continue
31
32             local = parts[4]
33             remote = parts[5]
34
35             try:
36                 local_addr, local_port = local.rsplit(":", 1)
37                 remote_addr, remote_port = remote.rsplit(":", 1)
38             except ValueError:
39                 continue
40
41             if remote_addr in ("127.0.0.1", "::1", "0.0.0.0", "*"):
42                 continue
43
44             if local_port == "1337" or remote_port == "*":
45                 continue
46
47             pid_only = "N/A"
48             if len(parts) > 6:
49                 raw_process = parts[-1]
50                 match = re.search(r"pid=(\d+)", raw_process)
51                 if match:
52                     pid_only = match.group(1)
53
54             connections.append({
55                 "LocalAddress": local_addr,
56                 "LocalPort": local_port,
57                 "RemoteAddress": remote_addr,
```

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```
58         "RemotePort": remote_port,
59         "State": parts[0],
60         "OwningProcess": pid_only
61     })
62
63     return connections, None
64
65 except Exception as e:
66     return None, str(e)
67
68
69 def monitor_tcp_connections():
70     while True:
71         try:
72             data, error = get_tcp_connections()
73
74             if error:
75                 socketio.emit("tcp_data", {"success": False, "error": error})
76                 socketio.sleep(2)
77                 continue
78
79             now = datetime.now()
80             current_time = (
81                 now.strftime("%Y-%m-%d %I:%M:%S")
82                 + f":{int(now.microsecond / 1000):03d}"
83             )
84
85             for conn in data:
86                 key = (
87                     conn["LocalAddress"],
88                     conn["LocalPort"],
89                     conn["RemoteAddress"],
90                     conn["RemotePort"],
91                     conn["OwningProcess"],
92                 )
93
94                 if key not in connection_timestamps:
95                     connection_timestamps[key] = current_time
96
97                 conn["FirstSeen"] = connection_timestamps[key]
98
99             socketio.emit("tcp_data", {"success": True, "connections": data})
100
101 except Exception as e:
102     socketio.emit("tcp_data", {"success": False, "error": str(e)})
103
104     socketio.sleep(2)
105
106
107 @socketio.on("get_tcp_connections")
108 def handle_get_tcp_connections():
109     print("[TCP Monitor] Client connected")
110     socketio.emit("client_info")
```

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```
107 @socketio.on("get_tcp_connections")
108 def handle_get_tcp_connections():
109     print("[TCP Monitor] Client connected")
110     socketio.emit("client_info")
111
112
113 @socketio.on("save_remote_ips")
114 def handle_save_remote_ips(ip_list):
115     with current_app.app_context():
116         try:
117             for ip in ip_list:
118
119                 if ip in ("127.0.0.1", "::1"):
120                     continue
121
122                 try:
123                     ipaddress.ip_address(ip)
124                 except ValueError:
125                     print(f"[INVALID IP] Skipping: {ip}")
126                     continue
127
128                 exists = db.session.query(RemoteIP).filter_by(ip_address=ip).first()
129
130                 if not exists:
131                     db.session.add(RemoteIP(ip_address=ip))
132
133             db.session.commit()
134
135         except Exception as e:
136             print(f"[DB ERROR] Failed to save remote IPs: {e}")
137
```

main/routes/homeRoute.py

```
1  from flask import Blueprint, render_template, session, redirect, url_for
2
3  homeRoute_bp = Blueprint("homeRoute_bp", __name__)
4
5
6  @homeRoute_bp.route("/board", methods=["GET", "POST"])
7  @homeRoute_bp.route("/rules", methods=["GET"])
8  @homeRoute_bp.route("/monitor", methods=["GET"])
9  @homeRoute_bp.route("/connection", methods=["GET"])
10 @homeRoute_bp.route("/sysinfo", methods=["GET"])
11 @homeRoute_bp.route("/about", methods=["GET"])
12 @homeRoute_bp.route("/profile", methods=["GET", "POST"])
13 @homeRoute_bp.route("/ssh")
14 def show_dashboard():
15     if session.get("logged_in"):
16         return render_template("index.html")
17     else:
18         return redirect(url_for("auth_bp.get_signin_page")), 302
19
```


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Threathunt.service

```
1 [Unit]
2 Description=Gunicorn instance to serve ThreatHunt
3 After=network.target
4
5 [Service]
6 User=root
7 Group=root
8 WorkingDirectory=/root/threathunt
9
10 Environment="PATH=/root/threathunt/venv/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"
11 Environment="PYTHONPATH=/root/threathunt"
12 Environment="PYTHONUNBUFFERED=1"
13
14 ExecStart=/root/threathunt/venv/bin/gunicorn \
15     --workers 1 \
16     --worker-class geventwebsocket.gunicorn.workers.GeventWebsocketWorker \
17     --bind 0.0.0.0:1337 \
18     --access-logfile /root/threathunt/access.log \
19     --error-logfile /root/threathunt/error.log \
20     --access-logformat '%(h)s %(l)s %(u)s %(t)s "%(r)s" %(s)s %(b)s "%(a)s"' \
21     --capture-output \
22     --enable-stdio-inheritance \
23     app:app
24
25 Restart=always
26 RestartSec=5
27
28 [Install]
29 WantedBy=multi-user.target
```


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Chapter V

Conclusion and Recommendation

Conclusion

The developed system effectively meets its objectives of detecting SSH intrusions, monitoring network activity, and providing users with tools to enhance network security. During testing, it successfully identified multiple failed login attempts, monitored active TCP connections, and allowed users to create customized firewall rules to block unauthorized access. The system's design prioritizes usability, making it accessible to both technical and non-technical users. Although the system has certain limitations—such as focusing solely on Windows platforms and specific threat types—it demonstrates strong potential for real-world application in network monitoring and protection. Overall, it contributes to creating a more secure digital environment by combining automated detection, user-driven configuration, and actionable insights.

Recommendation

For future development, it is recommended to expand the system's compatibility to include other operating systems such as Linux and macOS to accommodate a wider range of users. Adding support for additional threat types, such as malware detection and phishing attempts, would further enhance its capabilities. It is also suggested to integrate a notification or alert system to inform users in real time when potential intrusions or suspicious activities are

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detected. Moreover, including predefined firewall rule templates and geolocation-based IP analysis could help users respond more effectively to threats. Regular updates and machine learning-based threat prediction could also be introduced to strengthen the system's adaptability against emerging cyberattacks.

Final Thoughts

The project successfully delivers a functional and efficient solution for network security monitoring and intrusion detection. It bridges the gap between complexity and usability, offering a system that is both educational and practical. By empowering users to understand and manage their network security, it promotes awareness and proactive defense against cyber threats. With continued refinement and feature expansion, this system can evolve into a comprehensive cybersecurity tool that meets both academic and professional needs, supporting organizations in safeguarding their digital assets and maintaining a secure computing environment.

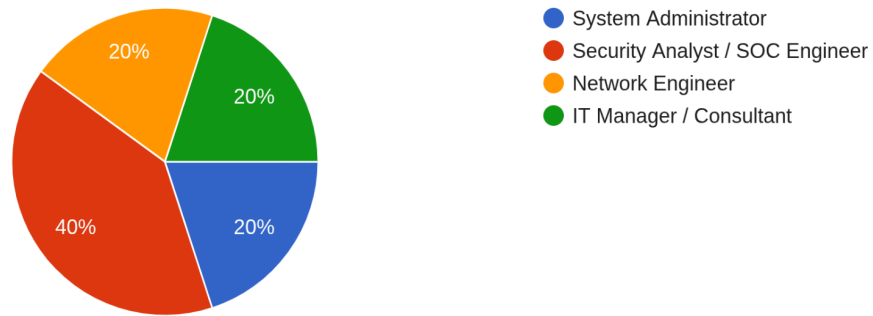
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Appendices

Appendix: User Survey Form and Summary

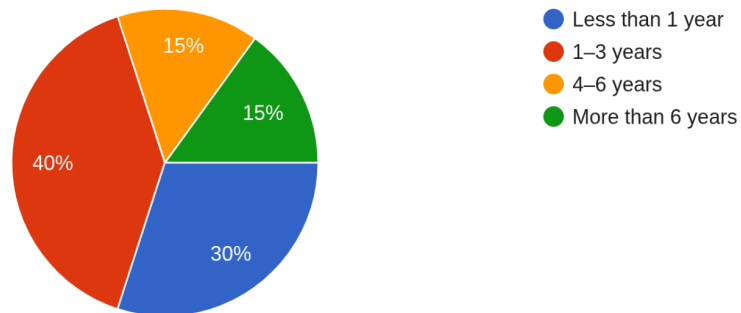
What is your primary role in the IT organization?

20 responses



How many years of experience do you have in network security or server management?

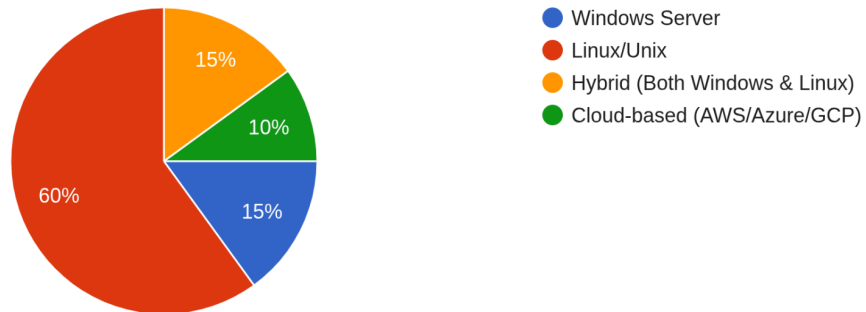
20 responses



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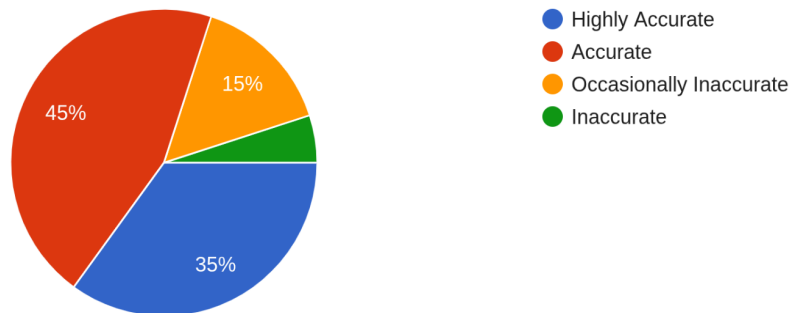
Which operating environment do you primarily manage?

20 responses



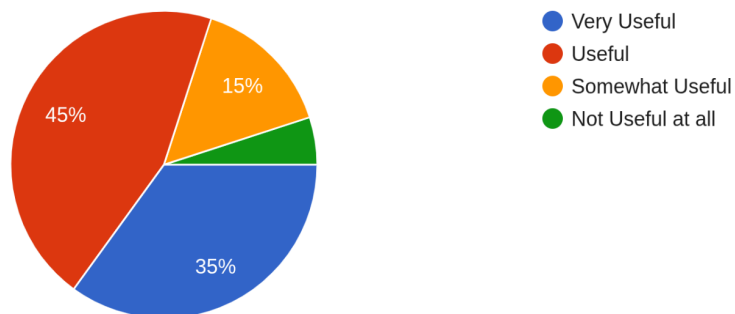
How accurately does the system detect failed SSH login attempts?

20 responses



How useful is the categorization of "Brute-Force Attacks" for your daily monitoring?

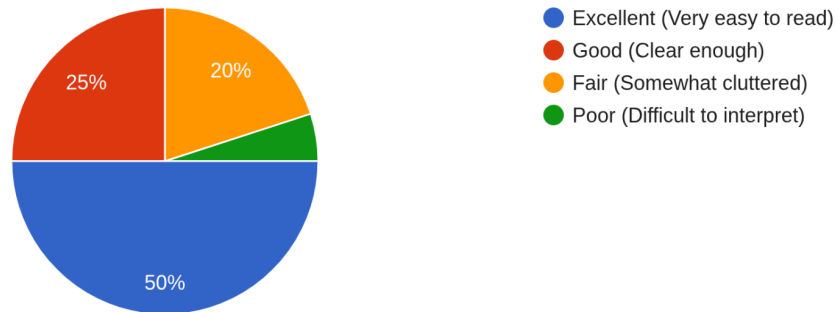
20 responses



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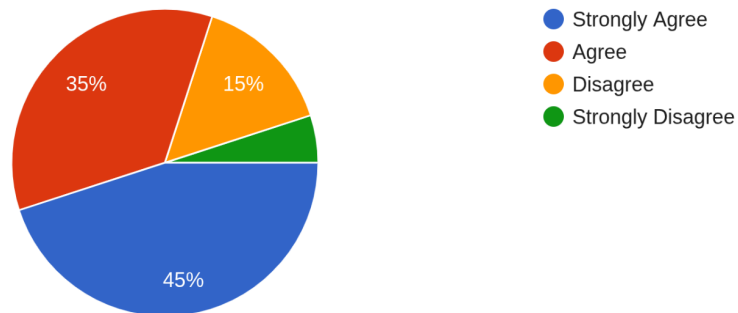
How would you rate the clarity of the SSH log entries (Timestamps, IPs, Usernames)?

20 responses



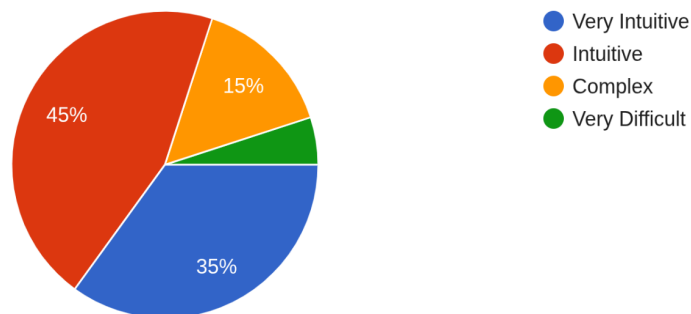
Does the system provide sufficient data for forensic analysis after a threat is detected?

20 responses



How intuitive is the process of creating new firewall rules in the system?

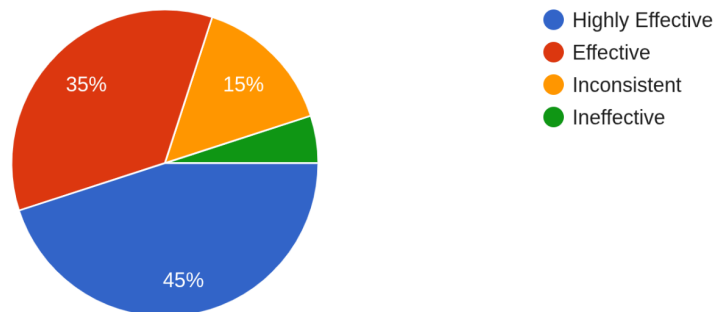
20 responses



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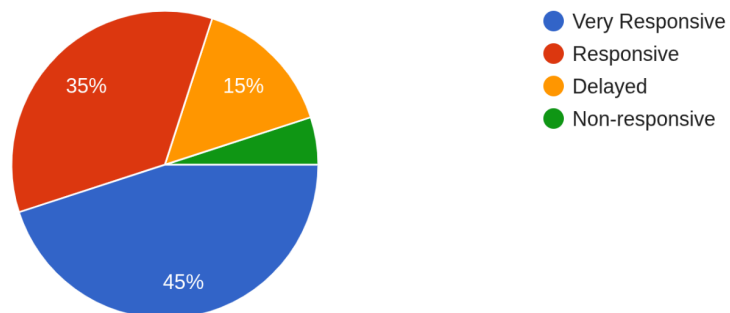
How effective is the system at blocking unauthorized access once a rule is set?

20 responses



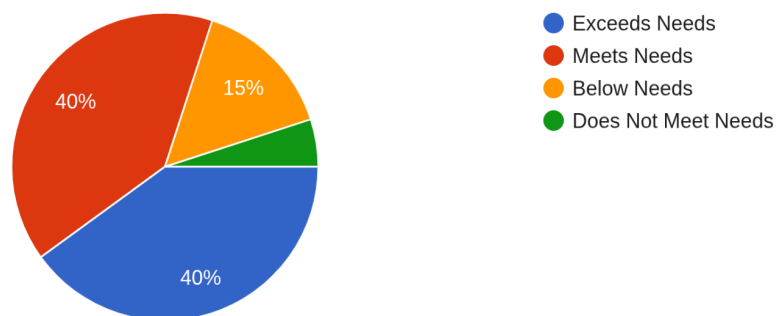
How would you rate the real-time display of active TCP connections?

20 responses



Does the monitoring of both listening and outgoing ports meet your security needs?

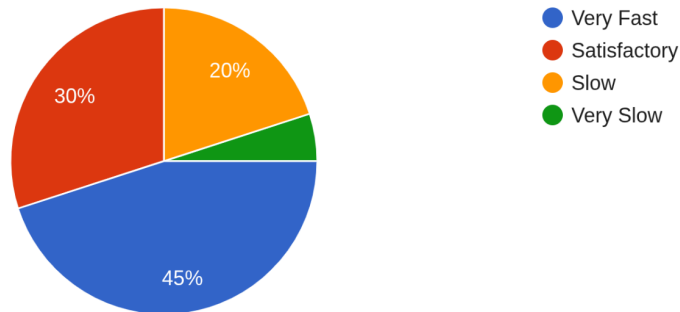
20 responses



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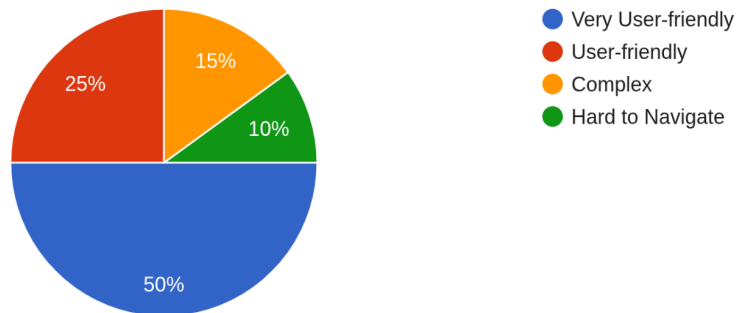
How would you rate the speed of the system when analyzing Windows Event logs?

20 responses



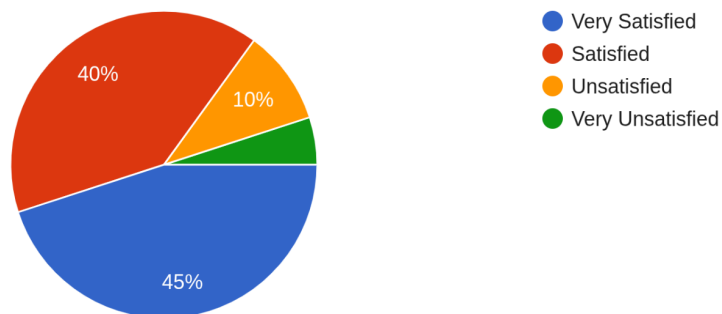
How user-friendly is the overall dashboard interface?

20 responses



How satisfied are you with the system's ability to handle multiple events simultaneously?

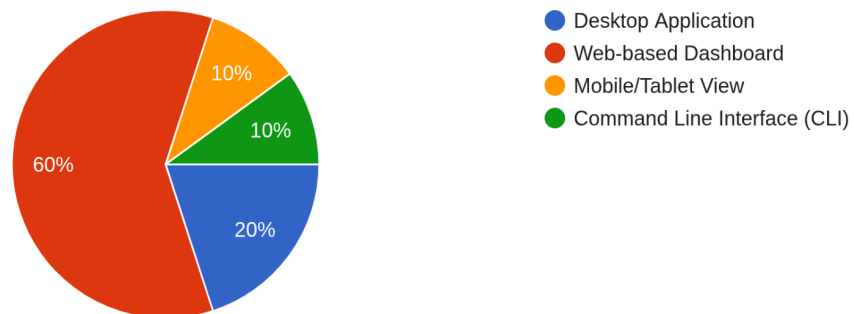
20 responses



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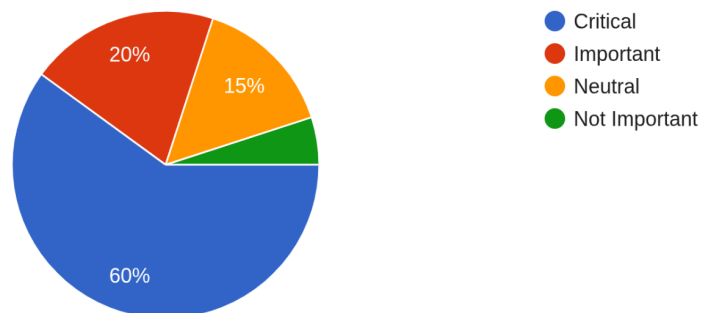
What is your preferred platform for monitoring these security alerts?

20 responses



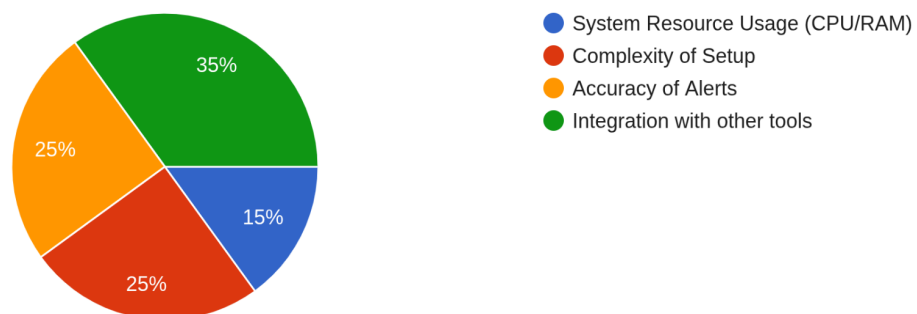
How important is the "Dynamic Management" (modifying rules on the fly) feature to you?

20 responses



What is your primary concern when using this intrusion detection system?

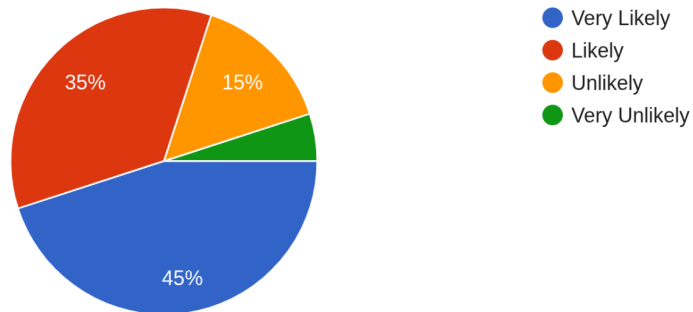
20 responses



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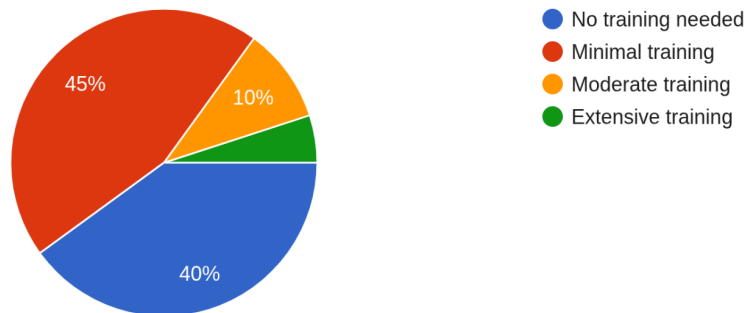
How likely are you to recommend ThreatHunt as a primary security tool?

20 responses



How much training do you feel is required to operate the Firewall module effectively?

20 responses



Overall, does the system successfully meet the objective of securing SSH and Network traffic?

20 responses

